

1 Configuration of OEE Key Figures

Purpose

This document describes the configuration options for customer-specific modifications of the Overall Equipment Effectiveness (OEE) and the respective KPIs:

- OEE
- Availability
- Performance
- Quality
- Net Equipment Effectiveness (NEE)
- Planned operating time
- Machine runtime
- Yield utilization
- Actual utilization



If you want to copy a formula from this document into the formula management using copy and paste, you must make sure that the formula does not include line breaks (CRLF). For this purpose, copy the formula into a text editor and remove the included line breaks before copying the formula into the formula management.

Requirements

The calculation of these key figures is based on the MDE log records. The key figures are included e.g. in the MOC application [OEE report](#).

Changes of the calculation rules are performed within [Formula management](#) or in the [INI configuration](#). They must be documented in the CID (Customer Implementation Documentation).

1.1 OEE Calculation

The Overall Equipment Effectiveness (OEE) is calculated as follows:

OEE = [Performance](#) x [Availability](#) x [Quality](#)

OEE is calculated as described below if no formula is defined for OEE within formula management:

$$oee = (rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11)) * (yield.primary / (yield.primary + scrap.primary + rework.primary + problem.primary)) * \text{performance_rate}$$

Technical standard formula to calculate OEE to be defined within [Formula management](#):

- Formula: oee

- Type: 5
- Calculation: $(rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11)) * (yield.primary / (yield.primary + scrap.primary + rework.primary + problem.primary)) * performance_rate$



If you want to change one component of the OEE calculation (that deviates from the standard formula), you **must** insert the complete formula for calculating OEE within formula management.

The OEE configuration must not include a formula name of any other key figure (e.g. avail, pf_rat, qual) in order to ensure a correct calculation. The only exception is the placeholder of the [performance_rate](#) because of its complex calculation. It is always the defined performance that is used here.

The formula name (oee) must be written in lower case letters.

1.2 Calculation of availability (productivity)

If formula management does not include a separate formula to calculate the key figure *availability*, the "availability" will be determined as part of the standard OEE calculation as follows:

$$Availability = \frac{\sum RPA_{11}}{\sum (RPA_{1-11})}$$

You can change the calculation rule for the availability in the configuration within [Formula management](#). The rule must be documented in the CID (Customer Implementation Documentation).

Technical standard formula to calculate the quality to be defined within HYDRA formula management:

- Formula: avail
- Type: 5
- Calculation: $rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11)$



If you have configured the formula *avail* within the formula management, you must also check or configure the formulas of the [OEE](#), [NEE](#), [Planned operating time](#), [Machine run time](#), [Yield utilization](#) and [Actual utilization](#). The formula name (avail) must be written in lower case letters.

1.3 Calculation of performance (effectiveness)

You can change the calculation rule for the performance in the configuration within [INI configuration](#) or [Formula management](#). It is included in the CID (Customer Implementation Documentation).

1.3.1 Standard as of MW 3.0

Version 1: Calculation based on recorded cycles

The performance is calculated based on the MDE log records. All data records collected during a production status (RPA11) are used. For this purpose, performance is calculated for *each* MDE log record in accordance with the formula

$$Performance = \frac{Target\ cycle}{Actual\ cycle} = \frac{Target\ cycle}{\left(\frac{RPA_{11}}{cycles}\right)}$$

The individual performance values are weighted in order to show a compressed view in evaluations. This weighting is based on the production time (RPA11).

$$Performance_{Product} = Performance * Duration_{RPA11}$$

$$Performance = \frac{Performance_{Product}}{\sum RPA_{11}}$$

Example of a performance calculation based on six MDE log records:

Machine	RPA11 [sec]	Performance	Performance (Product)
100	3600	0.7	2520
100	2700	0.8	2160
100	1800	0.9	1620
100	2700	0.5	1350
100	7200	0.9	6480
100	1800	0.9	1620
Total	19800	0.795	15750

Configuration in formula management

- The formula "pf_rat" must **not** exist in formula management.

Configuration in INI configuration

- No Ini configuration

Version 2: Calculation based on calculated cycles

If you do not want to use the recorded cycles of the MDE (cycles might not be recorded at all) for the calculation of the performance, you can use calculated cycles. In this case, you must configure as follows:

Configuration in formula management

- The formula "pf_rat" must **not** exist in formula management.

Configuration in INI configuration

Name	MDE
Section	PERFORMANCE_RATE
Key	STROKE_REPLACEMENT
Value	OFF
Active	☒

Version 3: Calculation based on recorded cycles and RPA 1-11

The performance is calculated based on the MDE log records. The following formula is used:

$$Performance = \frac{T \text{ arg et cycle}}{Actual \text{ cycle}} = \frac{T \text{ arg et cycle}}{\left(\frac{RPA_{1-11}}{cycles} \right)}$$

The individual performance values are weighted in order to show a compressed view in evaluations. This weighting is based on the total time (RPA1-11).

$$Performance_{Product} = Performance * Duration_{RPA1-11}$$

$$Performance = \frac{Performance_{Product}}{\sum RPA_{1-11}}$$

Configuration in formula management

- The formula "pf_rat" must **not** exist in formula management.

Configuration in INI configuration

Name	MDE
Section	PERFORMANCE_RATE
Key	CONSIDER_BMK01_11
Value	ON
Active	☒

Version 4: Calculation based on calculated cycles and RPA 1-11

The performance is calculated based on the MDE log records. The following formula is used:

$$Performance = \frac{Target\ cycle}{Actual\ cycle} = \frac{Actual\ cycle}{\left(\frac{RPA_{11}}{Cycles} \right)}$$

The individual performance values are weighted in order to show a compressed view in evaluations. This weighting is based on the total time (RPA1-11).

$$Performance_{Product} = Performance * Duration_{RPA1-11}$$

$$Performance = \frac{Performance_{Product}}{\sum RPA_{1-11}}$$

Configuration in formula management

The formula "pf_rat" must **not** exist in formula management.

Configuration in INI configuration

Make the following two entries:

Name	MDE
Section	PERFORMANCE_RATE
Key	STROKE_REPLACEMENT
Value	OFF
Active	☒

Name	MDE
Section	PERFORMANCE_RATE
Key	CONSIDER_BMK01_11
Value	ON
Active	☒

1.3.2 Compatibility up to and including MW 2.0

Up to and including MW 2.0, you must adjust the calculation of the performance. The following prerequisites are necessary. The formula must be as follows:



Required versions

- Java domain DomSvcProductionReporting Version 1.1.STD.8977
- Client domains
 - OeeReport 1.2.STD.20869
 - RPProductionReporting 1.2.STD.20870

Version 1: Old calculation based on recorded cycles

$$Performance = \frac{T \text{ arg et cycle}}{\left(\frac{\sum RPA11}{\sum Cycles} \right)} = \frac{\left(\frac{\sum (T \text{ arg et cycle} * RPA11)}{\sum RPA11} \right)}{\left(\frac{\sum RPA11}{\sum Cycles} \right)}$$

Configuration in formula management

- Formula: pf_rat
- Type: 5
- Calculation: cycle_target_weighted / cycle_actual_weighted

Configuration in INI configuration

Name	MDE
Section	PERFORMANCE_RATE
Key	STROKE_REPLACEMENT
Value	OFF
Active	☒



If you have configured the formula `pf_rat` within formula management, you must also configure the formula [oeo](#) for the OEE! To do so, you must replace the text `performance_rate` in the OEE formula with `cycle_target_weighted / cycle_actual_weighted`.

The formula name (`pf_rat`) must be written in lower case letters.

Version 2: Old calculation based on calculated cycles

$$Performance = \frac{T_{arg et cycle}}{\left(\frac{\sum RPA11}{\sum Cycles} \right)} = \frac{\left(\frac{\sum (T_{arg et cycle} * RPA11)}{\sum RPA11} \right)}{\left(\frac{\sum RPA11}{\sum Cycles} \right)}$$

Configuration in formula management

- Formula: `pf_rat`
- Type: 5
- Calculation: `cycle_target_weighted / cycle_actual_weighted`

Configuration in INI configuration

Ini entry: No Ini entry



If you have configured the formula `pf_rat` within formula management, you must also configure the formula [oeo](#) for the OEE! To do so, you must replace the text `performance_rate` in the OEE formula with `cycle_target_weighted / cycle_actual_weighted`.

You cannot weight different target cycle times. If you use this formula, it might happen that a performance >1 is calculated for an evaluation on group level, whereas a performance <1 is calculated for individual machines (which is then included in the evaluation).

The formula name (`pf_rat`) must be written in lower case letters.

1.3.3 Non-standard calculation according to formula

You can define calculation rules deviating from the standard using a customer-specific calculation. Please note: The MOC applications do not provide guarantee for the calculation of totals.

Configuration in formula management

- Formula: `pf_rat`

- Type: 5
- Calculation: requested calculation defined in formula management

Configuration in INI configuration

- Ini entry: No Ini entry



If you have configured the formula `pf_rat` within formula management, you must also configure the formula for the [OEE](#), [NEE](#) and the formula for the [Actual utilization](#). To do so, you must replace the text `performance_rate` in the OEE formula with the formula of `pf_rat`.

The formula name (`pf_rat`) must be written in lower case letters.

The acronyms mentioned below are available. In the formula, use the acronym before "="; the acronym behind "=" is the service acronym used in the calculation (e.g. in the OEE report).

```
rpa1=efficiencyreport.rpa1
rpa2=efficiencyreport.rpa2
rpa3=efficiencyreport.rpa3
rpa4=efficiencyreport.rpa4
rpa5=efficiencyreport.rpa5
rpa6=efficiencyreport.rpa6
rpa7=efficiencyreport.rpa7
rpa8=efficiencyreport.rpa8
rpa9=efficiencyreport.rpa9
rpa10=efficiencyreport.rpa10
rpa11=efficiencyreport.rpa11
rpa12=efficiencyreport.rpa12
yield.primary=efficiencyreport.yield.primary
scrap.primary=efficiencyreport.scrap.primary
rework.primary=efficiencyreport.rework.primary
problem.primary=efficiencyreport.problem.primary
cycle_actual_weighted=efficiencyreport.cycle_actual_weighted
cycle_target_weighted=efficiencyreport.cycle_target_weighted
```

The two values `cycle_target_weighted` and `cycle_actual_weighted` are specifically calculated acronyms:

$$\text{cycle_target_weighted} = \left(\frac{\sum (\text{Target cycle} * RPA11)}{\sum RPA11} \right)$$

$$\text{cycle_actual_weighted} = \left(\frac{\sum RPA11}{\sum \text{Cycles}} \right)$$

As of the above-mentioned versions, the following values/acronyms are additionally available:

$$\text{perf_rate_cycle_target} = \sum (\text{Target cycle} * RPA11)$$

$$\text{perf_rate_strokes} = \sum (\text{Cycles})$$

The values of these acronyms include (depending on the INI configuration) the recorded or calculated strokes.

1.4 Calculation of quality

If formula management does not include a separate formula to calculate the key figure *quality*, the "quality" will be determined as part of the standard OEE calculation as follows:

$$Quality = \frac{Yield_{primary}}{(Yield_{primary} + Scrap_{primary} + Rework_{primary} + Open\ Quantity_{primary})}$$

You can change the calculation rule for the quality in the configuration within [Formula management](#). The rule must be documented in the CID (Customer Implementation Documentation). Technical standard formula to calculate the quality:

- Formula: qual
- Type: 5
- Calculation: yield.primary / (yield.primary + scrap.primary + rework.primary + problem.primary)



If you have configured the formula *qual* within formula management, you must also check or configure the other formulas for [OEE](#), [NEE](#), [Yield utilization](#) and [Actual utilization](#).

The formula name (qual) must be written in lower case letters.

1.5 NEE Calculation

In contrast to the OEE, the Net Equipment Effectiveness (NEE) does not consider setup and configuration as a loss.

$$NEE = \frac{(RPA_{11} + RPA_7)}{\sum_{11} RPA} * Performance * Quality$$

NEE is calculated as described below if no formula is defined for NEE within formula management:

$$nee = ((rpa11 + rpa7) / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11)) * (yield.primary / (yield.primary + scrap.primary + rework.primary + problem.primary)) * \text{performance_rate}$$

Technical standard formula to calculate NEE to be defined within [Formula management](#):

- Formula: nee
- Type: 5

- Calculation: $((rpa11 + rpa7) / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11)) * (yield.primary / (yield.primary + scrap.primary + rework.primary + problem.primary)) * performance_rate$



If you want to change one component of the NEE calculation (that deviates from the standard formula), you **must** insert the complete formula for calculating NEE within formula management.

The NEE configuration must not include a formula name of any other key figure (e.g. avail, pf_rat, qual) in order to ensure a correct calculation. The only exception is the placeholder of the [performance_rate](#) because of its complex calculation. It is always the defined performance that is used here.

The formula name (nee) must be written in lower case letters.



The KPI *NEE* is only available if you enable the upgrade [oeerp82](#).

1.6 Planned operating time

Use the formula *op_ti* to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation:

Formula:

- Formula: op_ti
- Type: 5
- Calculation: $(rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11)$

1.7 Machine runtime

Use the formula *mch_rt* to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation:

- Formula: mch_rt
- Type: 5
- Calculation: rpa11

1.8 Yield utilization

Use the formula *yie_ut* to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation:

- Formula: *yie_ut*
- Type: 5
- Calculation: $\text{rpa11} * (\text{yield.primary} / (\text{yield.primary} + \text{scrap.primary} + \text{rework.primary} + \text{problem.primary}))$

1.9 Actual utilization

Use the formula *act_ut* to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Formula:

- Formula: *act_ut*
- Type: 5
- Calculation: $\text{rpa11} * (\text{yield.primary} / (\text{yield.primary} + \text{scrap.primary} + \text{rework.primary} + \text{problem.primary})) * \text{performance_rate}$



If you want to change one component of the calculation (that deviates from the standard formula), you **must** insert the complete formula *act_ut* for the calculation within formula management.

The configuration must not include a formula name of any other key figure (e.g. *avail*, *pf_rat*, *qual*) in order to ensure a correct calculation. The only exception is the placeholder of the [performance_rate](#) because of its complex calculation. It is always the defined performance that is used here.

If you have configured the formula *pf_rat* within formula management, you must also configure the formula *act_ut* for the actual utilization! To do so, you must replace the text *performance_rate* with the formula of *pf_rat*.

The formula name (here: *act_ut*) must be written in lower case letters.